

MD. AL AMIN CHY

Perspective Graduate Student

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WORK EXPERIENCE

1) Research Assistant

May '26 — Present

Dept. of Electrical and Electronic Engineering, Shahjalal University of Science and Technology Part-time, Bangladesh

- Applied FDTD, and TMM techniques to analyze optical responses, resonance behavior, sensing performance. [Link](#)
- Designed ultra-broadband metamaterial absorbers for solar energy harvesting and thin-film photovoltaic applications.

2) Research Assistant

Jan '25 — Jul '25

Faculty of Agriculture Engineering, Sylhet Agriculturer University Internship, Bangladesh

- Simulated using ANSYS HFSS and COMSOL Multiphysics, with hands-on experience in photonics labs. [Certificate](#)
- Conducted numerical analysis, RF/microwave simulations, and research manuscript preparation related to light matter interactions and electromagnetic structures.

PUBLICAITONS (6)

➔ **Md. Al Amin Chy**, M. A. H. Opu, M. F. Ahmed, R. Albab, M. JA Soeb. arXive. "Robust Optical Pulling of Plasmonic, Chiral, and Dielectric Nanorods in Single and Cluster Configurations Using Hyperbolic Metamaterials". Live

➔ **Md. Al Amin Chy**, TH Tasneem, SA Chowdhury, MRC Mahdy. arXive. "Ultra- Broadband Electromagnetic Control Using a Circular Ring Metasurface: Surface Wave Propagation, Beam Steering, and RCS Reduction (50 - 100 Ghz)". [Live](#)

➔ **Md. Al Amin Chy**, M.A.H. Opu, M.J.A. Soeb. ICERIE, 2025. Advance Engineering Research, part of Springer Nature "Optical Sorting of Distinct Nanorods Chiral Dielectric Hyperbolic Metamaterial in Clustered Configurations". [Live](#)

➔ **Md. Al Amin Chy**; Sabbir Ahmad Chowdhury; Rashedul Albab; Tasnimul Hasan Tasneem; Mizanur Rahman; Arif Ahammad. ICCIT(IEEE),2025, "Time Dependent Control of Voltage and Current via a Tunable Metasurface" [Live](#)

➔ M.F. Ahmad, Rashedul Albab, **Md. Al Amin Chy**, M.A.H. Sakib Opu, Tahsin Hadi, Tanvir Ahamed. ICCIT(IEEE), 2025, "Explainable Machine Learning for Multi-Class Power Quality Disturbance Classification Using SHAP and Feature Importance Analysis". [Live](#)

➔ M.F. Ahmad, M.A. Ahmad, **Md. Al Amin Chy**, S.N. Mehdi. ICCIT (IEEE), 2025, "Quantifying the Relationship between Socioeconomic Status and Parent-Child Attachment on Adolescents through Machine Learning". [Live](#)

RESEARCH EXPERIENCE

⇒ "A Multifunctional Metasurface Design: Achieving Superior Beam Steering, RCS Reduction, and Electromagnetic Wave Manipulation", Supervised by [Professor Dr. Mahdy Rahman Chowdhury](#) (Photonic Time Crystal, Metasurface) January 2024 - December 2024 | [Project](#)

- **Phase 1 (6 months)**: Studied fundamentals of electromagnetism; reproduced published results and compiled notes.
- **Phase 2 (6 months)**: First author on a journal manuscript; designed and optimized a multifunctional metasurface in CST (100+ simulations) and prepared figures using MATLAB and BLENDER.
- **Research Recognition**: Received a [supervisor-certified recognition letter](#) for successfully completing and submitting the research project for Q1 journal publication consideration under his supervision. .

⇒ "Robust Optical Pulling of Plasmonic, Chiral, and Dielectric Nanorods via Hyperbolic Metamaterials", Supervised by [Assistant Professor Md. Janibul Alam Soeb](#) (Optical Manipulation, Metamaterial, Plasmonics) January 2025 - June 2025 | [Project](#)

- **Conceptualization & Simulation (3.5 months)**: Studied relevant papers for idea development and physics understanding, reproduced published results, selected suitable materials, designed optical setups, and performed force calculations using COMSOL Multiphysics.
- **Scientific Paper Writing (2.5 months)**: First author on the manuscript; contributed to methodology, simulation analysis, results interpretation, and scientific writing under him

EDUCATION

B.Sc. (Engineering) in Electrical and Electronics Engineering, Shahjalal University of Science and Technology, (Final GPA: 3.88/4.0 with distinction) Jan '20 — Jul '25
Sylhet , Bangladesh

- Advanced Calculus, Differential Equations, Linear Algebra, Complex Variables, Embedded systems.
- Signals Processing, Analog & Digital Electronics, Electromagnetics, Quantum Mechanics, Solid State Device, RF & microwave, Materials, Communication, Renewable Energy, Power electronics. [Website](#)

SKILLS

CST Microwave Studio, COMSOL Multiphysics, ANSYS HFSS, Python, MATLAB, Microwave devices and passive components, RF circuit analysis and design, LaTeX, C++, Microsoft Excel, Powerpoint, Teamwork, Lumerical (FDTD)

PROJECTS

- 1) Three-Phase Inverter System using IGBT and Arduino** '24
▪ Designed DC–AC inverter with IGBT SPWM control for variable-frequency motor drives.
- 2) Line Following Robot**, Technoventure 2.0 Robotics Competition [Link](#) Apr '23 — Jul '23
▪ Built and programmed a sensor-based autonomous robot for real-time path detection and motion control.
- 3) Feasibility Study of a Renewable Energy Project using SAM**, Renewable Energy System Lab '24
▪ Conducted technical and financial assessment of solar PV and wind systems using SAM.

AWARDS & ACHIEVEMENTS

- 1) Academic Distinction Award** Jul '25
Shahjalal University of Science and Technology, I am proud to be ranked among the top 3% among others. [Link](#)
- 2) Merit-Based Scholarship (8 times)** Jun '25
Shahjalal University of Science and Technology, Consistently received throughout all semesters of my undergraduate program. [Link](#)
- 3) Best Campus Ambassador** Jan '24
Bangladesh Environment Olympiad 2024. Awarded for environmental awareness outreach. [Link](#)

CO-CURRICULAR ACTIVITIES

- 1) Organizing Secretary**, Aowlasher Social Welfare Organization (Govt.-registered, Reg. No. 869). Responsible for coordinating organizational activities. [Link](#) (October 2021 – September 2023.)
- 2) Publicity Secretary**, Electrical and Electronic Engineering Association. Responsible for publicity campaigns and communication. [Link](#) (March 2024 – June 2025)
- 3) Champion (2 times)**, Tournaments: 1) Engineer Football League, 2) Estrada: The Glory Indoor Football. Played as Left Forward. [Link](#) (December 2022 — December 2023)

REFERENCES

Md. Salah Uddin

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Dept. of Electrical and Electronics Engineering
Sylhet Engineering College, Sylhet, Bangladesh
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Arif Ahammad

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